

Randomization Mechanism Based on Elliptical Distributions for Location Privacy in Geo-Indistinguishability

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I. INTRODUCTION

Abstract—Geo-indistinguishability (GI), as a formal notion of location privacy, is implemented by perturbing the users' exact locations using random noise. For location-sensitive scenarios, e.g., localization and navigation, the introduction of random noise would decrease the data utility. In this paper, we investigate alternative randomization mechanisms that can provide higher data utility than the commonly used planar Laplacian mechanism while keeping the same privacy performance. Specifically, we investigate the usage of elliptical distributions to increase data utility. Firstly, we derive two sufficient and necessary conditions to determine whether a randomization mechanism satisfies GI. Using these conditions, we verify the feasibility of elliptical distributions for achieving GI and find that, except for the planar Laplacian, the Student's t distribution can also satisfy GI. Secondly, we adopt the Student's t distribution into the randomization mechanism to achieve GI and quantify its performance in terms of privacy level and data utility. To optimize the data utility and privacy level of Student's t distribution in practical applications, we formulate two optimization problems: one aims to maximize data utility under a given privacy level, and the other aims to maximize the privacy guarantee under the constraint of data utility. Extensive experiments have been conducted in two typical location-based applications, including fingerprint-based indoor localization and location-based advertising. Both theoretical analysis and experimental results evaluate the performance of the Student's t distribution and provide insight into the selection of randomization mechanism under different scenarios by comparing Student's t mechanism with the Laplacian one.

Index Terms—Geo-indistinguishability, elliptical distributions, student's t distribution, and location privacy.

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LOCATION-BASED services (LBS) have found extensive applications in daily life, such as points of interest recommendation, navigation, and surveillance [1]. The quality of LBS highly depends on accurate location information. However, sharing locations with LBS providers or untrusted third parties can lead to disclosing sensitive information, e.g., home address, daily habits, and economic conditions [2], [3], [4]. Adversaries can take advantage of this disclosed location information to reconstruct users' trajectories using some fine-grained inference methods [5] or speed-based trajectory inference framework [6]. Therefore, preserving location privacy is critical in LBS.

The existing location privacy preservation technologies can be classified into three categories: noise-adding methods [5], [7], [8], [9], [10], information-hiding methods [11], [12], [13], [14] and encryption-based methods [15], [16], [17], [18]. Among these privacy preservation techniques, noise-adding methods have gained extensive interest due to their high effectiveness and low computational complexity. One of the most common notions defining the location privacy level of noise-adding methods is geo-indistinguishability (GI), which can be regarded as a form of differential privacy (DP) in Euclidean metric [10].

The implementation of GI relies on perturbing actual locations using a randomization mechanism. While GI provides strong guarantees for location privacy, it also degrades data utility and system performance. Existing research on GI mainly falls into two categories. The first category studies the application of the original GI notion in specific scenarios, such as localization systems [4], [19], [20], spatial crowdsourcing [21], [22], and ride-sharing services [23], [24]. In these works, the planar Laplacian mechanism remains the most commonly used mechanism to implement GI. The second category extends GI by redefining the privacy notion for specific applications, aiming to improve utility while preserving location privacy [25], [26], [27], [28], [29]. These GI variants are typically implemented through tailored mechanisms related to the exponential mechanism and rely on application specific perturbation designs. In the GI framework, the exponential mechanism can be regarded as a discrete planar Laplacian mechanism [30]. However, little work has explored alternative randomization mechanisms for the original GI definition. Our contribution lies in investigating fundamental randomization mechanisms for achieving GI. Our proposed mechanisms

not only achieve much better performance than the planar Laplacian distribution while maintaining broad applicability, but also provide useful insights for mechanism design in GI variants.

Since the planar Laplacian distribution is a special case of elliptical distributions, it is natural to explore general elliptical distributions for GI. However, it remains unclear whether these elliptical distributions meet GI criteria. It is quite difficult to directly check whether a privacy mechanism satisfies GI by the original definition. Moreover, when adopting an elliptical distribution into the randomization mechanism for GI, we need to determine the proper value of key parameters in the feasible elliptical distribution to achieve a certain privacy level. Therefore, as the fundamental guideline of parameter design, the quantitative relationship between privacy level and parameters of the noise distribution should be established. For a feasible randomization mechanism for GI, how to determine the corresponding privacy level under GI definition is also challenging. Besides, the design of the randomization mechanism should jointly consider the privacy level and the data utility. In practical applications, it is also challenging to optimize the feasible randomization mechanisms by taking both the privacy level and data utility into consideration.

Motivated by the above discussions, in this work, we investigate the feasibility of elliptical distributions for GI, and the quantitative relationship between privacy level and key parameters of the feasible elliptical distributions. Then we consider the optimal design problem of randomization mechanisms that utilizes feasible elliptical distributions for different cost functions with respect to the location privacy and data utility. The main contributions of this work are summarized as follows:

- 1) To the best of our knowledge, this is the first work that investigates the use of elliptical distributions in the randomization mechanism to achieve GI. We derive a necessary and sufficient condition for GI, which is then used to analyze the feasibility of elliptical distributions for GI. We find that, except for planar Laplacian distribution, another typical elliptical distribution, Student's t distribution is also able to ensure GI.
- 2) Before designing the randomization mechanism under the Student's t distribution, we analyze the privacy level and data utility of this distribution. This analysis can be used to optimize the performance regarding location privacy and data utility under the randomization mechanism with Student's t distribution.
- 3) To balance the data utility and privacy level of Student's t distribution, we formulate two problems to optimize the randomization mechanism under Student's t distribution in different scenarios. One aims to maximize data utility while maintaining a given privacy level, and the other aims to minimize the privacy level subject to the constraint of data utility. Closed-form solutions are provided for both problems.

The remainder of this paper is organized as follows. Section II presents the related works. In Section III, we introduce the background on GI and present the problem formulation. Next, in Section IV, we analyze the feasibility of elliptical distributions for achieving GI and verify that Student's t distribution can also

ensure GI. Section V establishes the quantitative relationship between privacy level and parameters of Student's t distribution. In Section VI, we design the optimal randomization mechanism under Student's t distribution by formulating two optimization problems. Section VII evaluates the performance of the proposed mechanism in two applications. Section VIII discusses the findings and summarizes future works. Finally, Section IX concludes this paper.

Notations: $\mathbb{E}[\cdot]$ denotes the expectation operator. Each vector is denoted by lower-case bold letters, and each matrix is denoted by upper-case bold letters. The trace, determinant, transpose, and inverse of a matrix are denoted by $\text{tr}(\cdot)$, $\det(\cdot)$, $(\cdot)^T$ and $(\cdot)^{-1}$, respectively. The L_2 norm of a vector is denoted by $\|\cdot\|_2$.

II. RELATED WORK

A. Differential Privacy

DP is a well-known notion for protecting individual privacy in a database while providing useful statistical information [31]. It has been applied in various applications, such as Google Chrome browsers [32], 2020 US Census [33], and machine learning schemes [34], [35], [36]. The wide applications of DP can be attributed to its simple implementation and strong privacy guarantee. DP adds noise to data to protect the privacy of individuals, which would reduce the utility of the data. Existing research on DP mainly focuses on developing private algorithms to balance privacy and utility.

Since the advent of DP, various randomization mechanisms have been developed to ensure its implementation. The Laplacian and Gaussian mechanisms are the most widely used mechanisms for achieving DP [31]. He et al. investigated the conditions of differential privacy in the single-dimensional case and then applied these conditions to analyze the privacy-preserving consensus algorithm [37]. Geng et al. derived the optimal DP mechanism by formulating a general utility-maximization problem [38]. They demonstrated that the optimal DP mechanism follows a staircase-shaped probability density function (PDF). Theoretical analysis verified that the utility of the staircase mechanism is larger than the Laplacian mechanism when the privacy requirement is not very high.

Sadeghi et al. proposed offset-symmetric Gaussian distribution (OSGT) for achieving DP, which is obtained using the normalized tails of two symmetric Gaussians around zero [39]. They demonstrated that the OSGT mechanism offers better privacy-utility performance than the Gaussian and Laplacian mechanisms. Muthukrishnan and Kalyani introduced the flipped Huber mechanism to achieve (ϵ, δ) -DP [40]. This mechanism combines Laplace-like behavior near the mean with Gaussian-like tails, offering a better trade-off between privacy and utility compared to pure Gaussian and Laplace mechanisms. Jiang et al. proposed a hybrid differential privacy mechanism that combines Gaussian and Laplace perturbations within federated learning to enhance the privacy-preserving performance of medical image classification [41].

The above works investigate univariate randomization mechanisms for achieving DP, while some works investigate the use of multivariate elliptical distributions as randomization

mechanisms for DP. Hardt and Talwar proposed the K -Norm mechanism, a type of elliptical distribution that leverages geometric insights to provide nearly optimal error bounds for DP in answering linear queries [42]. Existing DP mechanisms are widely applied in scenarios where aggregated results about multiple users are published. However, they are generally unsuitable for applications involving individual data, where data utility is highly sensitive to perturbations.

B. Geo-Indistinguishability

In the context of protecting location privacy, location information is often highly sensitive. Andrés et al. introduced the notion of GI by extending DP into Euclidean distance [10]. The theoretical robustness of extending DP to GI relies on the well-established framework that generalizes DP from the Hamming distance to more general metrics. Current studies have shown that DP can also be generalized using other metrics depending on the structure of the data to be protected [43] [44], such as Earth Mover's Distance for user-level multiple record data [45], temporal distance for privacy-preserving trajectory publishing [46], and Word Mover's Distance for natural language processing [47]. GI considers the distance between locations in addition to the privacy budget, as location privacy is sensitive to the distance between the actual location and a perturbed location. To achieve GI, Andrés et al. presented a randomization mechanism that adds random noise, drawn from a planar Laplacian distribution, to the user's location.

The formal privacy guarantee provided by GI has supported its use in a variety of application scenarios. In localization systems, GI has been used to protect the location privacy of cooperative nodes during localization [4], [20], and to preserve users' privacy in indoor fingerprinting localization by locally obfuscating RSS vectors before reporting [19]. In spatial crowdsourcing, GI has been applied to protect workers' reported locations during worker recruitment and task allocation [21], [22]. In ride-sharing services, GI has been used to generate perturbed locations for privacy-preserving matching [23], [24]. In most of these works, the planar Laplacian distribution is adopted as the standard randomization mechanism for achieving GI. However, the introduction of random noise inevitably decreases the data utility. Motivated by this, our work investigates the feasibility of using elliptical mechanisms to achieve GI, aiming to improve data utility while keeping the same privacy performance.

To improve data utility while ensuring location privacy, existing works have redefined GI for different applications. Zhao et al. proposed geo-ellipse-indistinguishability for directional distribution analysis by extending GI into Mahalanobis space [25]. The proposed elliptical obfuscation mechanism improves utility compared to the planar Laplacian mechanism, using utility metrics like Frobenius distance and Riemannian distance between the covariance matrices of actual and perturbed locations. Ren et al. proposed DistPreserv to preserve users' location distributions while ensuring location privacy in scenarios where maintaining statistical location distributions is crucial, such as traffic congestion analysis [26]. Its utility is measured by the Jensen-Shannon divergence, which evaluates

the similarity between the actual and perturbed location distributions. Compared to the planar Laplacian mechanism, it better retains the original location distribution. Takagi et al. introduced geo-graph-indistinguishability (GeoGI) for road network, and measures utility as the expected shortest path distance between a user's actual and perturbed locations [27]. The proposed graph-exponential mechanism for GeoGI, evaluated using the expected shortest path distance between actual and perturbed locations, achieves higher utility on road networks compared to the planar Laplacian mechanism under the same privacy level.

Although variants of GI can improve utility in specific scenarios, they are unsuitable for many other LBS applications. For example, geo-ellipse-indistinguishability is designed for directional distribution analysis but is inapplicable to scenarios where the required services do not rely on location distribution, such as k -nearest neighbor search [25]. GeoGI utilizes road network knowledge to enhance privacy guarantee but is inapplicable to scenarios without road network, such as indoor localization [27]. Fundamental randomization mechanisms for the original GI definition are of broader applicability and scalability. However, there are few alternative randomization mechanisms that can achieve pure GI while providing higher data utility than the planar Laplacian mechanism.

III. PRELIMINARIES AND PROBLEM FORMULATION

A. Differential Privacy and Geo-Indistinguishability

DP was initially proposed to protect the privacy of sensitive information within databases [31]. It can provide a strong privacy guarantee statistically without revealing individuals' sensitive information in the dataset. A privacy mechanism is an algorithm that takes a database, a set of possible data entries, random bits, and a set of queries as input, producing outputs that ensure privacy by introducing randomness. To achieve DP, a privacy mechanism can be implemented in various ways, such as adding noise, and randomized response. Let $\mathcal{M}$ be a randomization mechanism and $\text{Range}(\mathcal{M})$ denote the set of all output values it may produce. The definition of DP, as described by Dwork [31], is as follows:

Definition 1: $\mathcal{M}$ is ϵ -differentially private if for all subsets $S \subseteq \text{Range}(\mathcal{M})$ and for all databases D, D' satisfying $\|D - D'\|_1 \leq 1$, the following inequality holds:

$$\frac{P[\mathcal{M}(D) \in S]}{P[\mathcal{M}(D') \in S]} \leq e^\epsilon, \quad (1)$$

where $\|D - D'\|_1$ is the Hamming distance between two databases D and D' , and it represents the number of elements in which D and D' differ. The parameter $\epsilon > 0$ denotes the level of privacy. A smaller ϵ corresponds to a stronger privacy guarantee.

The notion of DP can be generalized by making the distance metric no longer limited to Hamming distance. In the context of location privacy, Andrés et al. introduced the notion of GI [10]. They extended the concept of DP by using the Euclidean distance as the metric for measuring the distance between possible inputs. Let $\mathcal{K}$ be a randomization mechanism and $\text{Range}(\mathcal{K})$ denote the set of possible reported spatial points. The definition of GI is as follows.

TABLE I
LIST OF THE TYPICAL ELLIPTICAL DISTRIBUTIONS

Elliptical distributions	PDF	Covariance
Planar Laplacian	$f_{\mathcal{L}}(\mathbf{z} \mathbf{x}) = \frac{\det(\mathbf{\Sigma})^{-\frac{1}{2}}}{2\pi} e^{-\sqrt{(\mathbf{z}-\mathbf{x})^T \mathbf{\Sigma}^{-1}(\mathbf{z}-\mathbf{x})}}$	$3\mathbf{\Sigma}$
Gaussian	$f_{\mathcal{N}}(\mathbf{z} \mathbf{x}) = \frac{\det(\mathbf{\Sigma})^{-\frac{1}{2}}}{2\pi} e^{-\frac{1}{2}(\mathbf{z}-\mathbf{x})^T \mathbf{\Sigma}^{-1}(\mathbf{z}-\mathbf{x})}$	$\mathbf{\Sigma}$
Student's t ¹	$f_{\mathcal{T}}(\mathbf{z} \mathbf{x}) = \frac{\det(\mathbf{\Sigma})^{-\frac{1}{2}}}{2\pi} \left[1 + \frac{1}{v}(\mathbf{z}-\mathbf{x})^T \mathbf{\Sigma}^{-1}(\mathbf{z}-\mathbf{x}) \right]^{-\frac{v+2}{2}}$	$\frac{v-2}{v} \mathbf{\Sigma}$
Logistic ²	$f_{\mathcal{C}}(\mathbf{z} \mathbf{x}) = \frac{\det(\mathbf{\Sigma})^{-\frac{1}{2}}}{2\pi} \frac{e^{-\frac{1}{2}(\mathbf{z}-\mathbf{x})^T \mathbf{\Sigma}^{-1}(\mathbf{z}-\mathbf{x})}}{[1 + e^{-\frac{1}{2}(\mathbf{z}-\mathbf{x})^T \mathbf{\Sigma}^{-1}(\mathbf{z}-\mathbf{x})}]^2}$	-

Definition 2: A mechanism $\mathcal{K}$ satisfies ϵ -GI iff for all observations $S \subseteq \text{Range}(\mathcal{K})$:

$$\frac{P(S|\mathbf{x})}{P(S|\mathbf{x}')} \leq e^{\epsilon r}, \quad \forall r > 0 \quad \forall \mathbf{x}, \mathbf{x}' : \|\mathbf{x} - \mathbf{x}'\|_2 \leq r, \quad (2)$$

where $P(S|\mathbf{x})$ and $P(S|\mathbf{x}')$ are the probabilities to generate locations in S respectively at location $\mathbf{x}$ and $\mathbf{x}'$, $\|\mathbf{x} - \mathbf{x}'\|_2$ denotes the Euclidean distance between $\mathbf{x}$ and $\mathbf{x}'$, and r is the maximum distance between $\mathbf{x}$ and $\mathbf{x}'$.

This definition illustrates that the ratio of probabilities generating the same perturbed locations at $\mathbf{x}$ and $\mathbf{x}'$ respectively is bounded by $e^{\epsilon r}$. When ϵr is small enough, based on the perturbed location points in S , it is difficult to tell whether the actual location is $\mathbf{x}$ or $\mathbf{x}'$. This definition generally defines the privacy level by $\ell = \epsilon r$. The smaller ℓ is, the stronger the privacy guarantee the user enjoys within r .

GI can be viewed as an extension of DP under the Euclidean metric. Specifically, two adjacent databases D and D' in DP correspond to two locations $\mathbf{x}$ and $\mathbf{x}'$ in GI. By extending the notion of adjacency from differing by a Hamming distance of 1 to a Euclidean distance of r , the standard definition of DP can be extended to GI. Despite this connection, there are key differences between DP and GI. Firstly, DP and GI adopt different metrics for the private data. DP requires the Hamming distance between two databases to be less than or equal to 1, while GI requires the maximum Euclidean distance between two actual locations to be r . Secondly, the privacy levels of DP and GI are defined differently. The privacy budget of DP, denoted as ϵ , reflects the privacy level. For GI, the privacy level is distance-dependent, given by $\ell = \epsilon r$. Since DP and GI are defined differently, the theoretical results and randomization mechanisms for DP can not directly apply to GI.

B. Problem Formulation

The planar Laplacian mechanism is commonly used for achieving GI, and it belongs to the class of elliptical distributions. In location and navigation scenarios, the introduction of Laplacian random noise often reduces data utility. A natural question is whether other elliptical distributions can also be employed to achieve GI while improving data utility. We are interested in the feasibility of elliptical distributions for GI.

For an elliptical distribution, if its PDF can be written in a closed-form expression, it generally follows a specific form

given by [48]:

$$f(\mathbf{z}|\mathbf{x}) = \frac{\Gamma(\frac{p}{2}) \det(\mathbf{\Sigma})^{-\frac{1}{2}}}{(2\pi)^{\frac{p}{2}}} g(d_{\mathbf{z},\mathbf{x}}^2), \quad (3)$$

where $\mathbf{z}$ is a p -dimensional random vector with a median vector $\mathbf{x}$, $\Gamma(\frac{p}{2})$ is the Gamma function, $d_{\mathbf{z},\mathbf{x}} = \sqrt{(\mathbf{z}-\mathbf{x})^T \mathbf{\Sigma}^{-1}(\mathbf{z}-\mathbf{x})}$ is the Mahalanobis distance between $\mathbf{z}$ and $\mathbf{x}$, $\mathbf{\Sigma}$ is a positive definite matrix that is proportional to the covariance matrix, and $g(\cdot)$ is a non-negative and monotonically decreasing function of $d_{\mathbf{z},\mathbf{x}}^2$. Table I presents the PDFs of several common elliptical distributions. We consider the location privacy protection problem in 2-dimensional space, i.e., $p = 2$, $\mathbf{x}$ represents the actual location and $\mathbf{z}$ is the perturbed location generated by the randomization mechanism.

In this paper, we investigate using elliptical distributions as alternative randomization mechanisms to ensure GI. The following issues are to be addressed:

First, the feasibility of elliptical distributions for GI is unclear. Therefore, it is essential to evaluate their feasibility for GI by investigating the fundamental necessary and sufficient conditions of GI. These conditions will enable the analysis of various classes of elliptical distributions to determine whether a distribution can be applied to ensure GI.

Secondly, when adopting an elliptical distribution in the randomization mechanism for GI, its performance in terms of privacy level and data utility is unknown. Regarding the privacy level, we will establish the quantitative relationship between the privacy level and the statistical parameters of the feasible distribution. To evaluate the data utility of randomization mechanisms, the statistical distance between the perturbed locations and the actual location is considered as a utility metric. Intuitively, a larger distance between the actual location and the perturbed location results in lower LBS quality, such as incorrect navigation route or irrelevant recommendation, thereby degrading the data utility. This metric has been adopted in the literature as a standard measure of utility [27], [49], [50], [51], [52], [53], [54], and it is a general utility metric independent of specific location-based applications. Specifically, the data utility

¹In Student's t distribution, $v > 2$ is the degree of freedom (DoF). When $0 < v \leq 2$, the covariance matrix intends to infinity. When v increases to infinity, the Student's t distribution becomes a Gaussian distribution.

²The covariance matrix of $\mathbf{z}$ generated from the Logistic distribution does not have a closed-form representation.

is defined as:

$$u = \frac{1}{\mathbb{E}[(\mathbf{z} - \mathbf{x})^T(\mathbf{z} - \mathbf{x})]}. \quad (4)$$

Thirdly, the performance of a randomization mechanism is closely associated with parameters of the statistical distribution. Finding parameters corresponding to the optimal performance is also a critical issue. To design the optimal randomization mechanism, we consider two different scenarios. In the first scenario, an optimization problem is formulated to maximize the data utility for a given privacy level. In the second scenario, an optimization problem is formulated to minimize the privacy level while maintaining the expected data utility.

IV. FEASIBILITY ANALYSIS OF ELLIPTICAL DISTRIBUTIONS FOR GI

In this section, we analyze the feasibility of elliptical distributions for GI. Firstly, we derive two fundamental necessary and sufficient conditions of GI. These conditions can be used to evaluate the feasibility of various randomization mechanisms. Then, we apply the obtained conditions to analyze the feasibility of four typical elliptical distributions, including planar Laplacian, Gaussian, Student's t , and Logistic distributions.

A. Necessary and Sufficient Conditions

It is quite difficult to directly check whether a privacy mechanism satisfies GI by Definition 2. Given a randomization mechanism $\mathcal{K}$, let $S \subseteq \text{Range}(\mathcal{K})$ be a set of possible reported spatial points. We provide the following necessary and sufficient conditions to determine whether or not it satisfies GI.

Lemma 1: A randomization mechanism $\mathcal{K}$ satisfies ϵ -GI if and only if, for all $S \subseteq \text{Range}(\mathcal{K})$, the following holds:

$$\frac{f(\mathbf{z}|\mathbf{x})}{f(\mathbf{z}|\mathbf{x}')} \leq e^{\epsilon r}, \quad \forall r > 0, \quad \forall \mathbf{x}, \mathbf{x}' : \|\mathbf{x} - \mathbf{x}'\|_2 \leq r, \quad (5)$$

for $\mathbf{z} \in S$ almost everywhere.

In (5), $f(\mathbf{z}|\mathbf{x})$ and $f(\mathbf{z}|\mathbf{x}')$ are the PDFs to generate the perturbed point $\mathbf{z}$ respectively from $\mathbf{x}$ and $\mathbf{x}'$.

Proof: Both the necessity and the sufficiency are proved.

1) *Necessity:* We prove the necessity by contradiction. Supposing there exists a non zero-measure set S_0 such that for any $\mathbf{z} \in S_0$, $\frac{f(\mathbf{z}|\mathbf{x})}{f(\mathbf{z}|\mathbf{x}')} > e^{\epsilon r}$. Consequently, we have

$$\frac{P(S_0|\mathbf{x})}{P(S_0|\mathbf{x}')} = \frac{\int_{S_0} f(\mathbf{z}|\mathbf{x}) d\mathbf{z}}{\int_{S_0} f(\mathbf{z}|\mathbf{x}') d\mathbf{z}} > \frac{e^{\epsilon r} \int_{S_0} f(\mathbf{z}|\mathbf{x}') d\mathbf{z}}{\int_{S_0} f(\mathbf{z}|\mathbf{x}') d\mathbf{z}} = e^{\epsilon r},$$

which contradicts with the definition of ϵ -GI and implies that $\mathcal{K}$ does not satisfy ϵ -GI. Therefore, the necessity is proved.

2) *Sufficiency:* Since (5) holds for all S , we have

$$\frac{P(S|\mathbf{x})}{P(S|\mathbf{x}')} = \frac{\int_S f(\mathbf{z}|\mathbf{x}) d\mathbf{z}}{\int_S f(\mathbf{z}|\mathbf{x}') d\mathbf{z}} \leq \frac{e^{\epsilon r} \int_S f(\mathbf{z}|\mathbf{x}') d\mathbf{z}}{\int_S f(\mathbf{z}|\mathbf{x}') d\mathbf{z}} = e^{\epsilon r}.$$

Therefore, (5) is a sufficient condition for ϵ -GI. $\square$

Remark 1: In mathematics, "almost everywhere" means that a property holds at all points except for a set of measure zero. While $\frac{f(\mathbf{z}|\mathbf{x})}{f(\mathbf{z}|\mathbf{x}')}$ might not be less than or equal to $e^{\epsilon r}$ on the zero measure set, this does not affect the GI property because the

integral over a zero measure set is zero. Therefore, if $\frac{f(\mathbf{z}|\mathbf{x})}{f(\mathbf{z}|\mathbf{x}')} \leq e^{\epsilon r}$ holds at all points except for those in a set of measure zero, i.e., the condition holds almost everywhere, the mechanism still satisfies ϵ -GI.

Although Lemma 1 provides fundamental conditions for determining whether a randomization mechanism satisfies ϵ -GI, it is still difficult to check if (5) holds in practical applications. For elliptical distributions with closed-form PDF, we can obtain other necessary and sufficient conditions, which is easy to evaluate the feasibility for ϵ -GI.

Theorem 1: A randomization mechanism $\mathcal{K}$ generated by an elliptical distribution with PDF $f(\mathbf{z}|\mathbf{x})$ satisfies ϵ -GI if and only if, for all $S \subseteq \text{Range}(\mathcal{K})$, the following holds:

$$\frac{g(d_{\mathbf{z},\mathbf{x}}^2)}{g[(d_{\mathbf{z},\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]} \leq e^{\epsilon r}, \quad \forall r > 0, \quad \forall \mathbf{x}, \mathbf{x}' : \|\mathbf{x} - \mathbf{x}'\|_2 \leq r,$$

for $\mathbf{z} \in S$ almost everywhere.

Proof: The key idea is to first upper bound $d_{\mathbf{z},\mathbf{x}'}^2$ under a fixed $d_{\mathbf{z},\mathbf{x}}^2$. Since the density $f(\mathbf{z}|\mathbf{x}')$ is determined by $g(d_{\mathbf{z},\mathbf{x}'}^2)$, the monotonicity of $g(\cdot)$ then yields an upper bound on the density ratio $\frac{f(\mathbf{z}|\mathbf{x})}{f(\mathbf{z}|\mathbf{x}')}$.

According to Lemma 1, to determine if an elliptical distribution satisfies ϵ -GI, we need to check whether the following ratio can be bounded by $e^{\epsilon r}$:

$$\frac{f(\mathbf{z}|\mathbf{x})}{f(\mathbf{z}|\mathbf{x}')} = \frac{\frac{\Gamma(\frac{p}{2})\det(\Sigma)^{-\frac{1}{2}}}{(2\pi)^{\frac{p}{2}}} g(d_{\mathbf{z},\mathbf{x}}^2)}{\frac{\Gamma(\frac{p}{2})\det(\Sigma)^{-\frac{1}{2}}}{(2\pi)^{\frac{p}{2}}} g(d_{\mathbf{z},\mathbf{x}'}^2)} = \frac{g(d_{\mathbf{z},\mathbf{x}}^2)}{g(d_{\mathbf{z},\mathbf{x}'}^2)}, \quad (6)$$

where $d_{\mathbf{z},\mathbf{x}'} = \sqrt{(\mathbf{z} - \mathbf{x}')^T \Sigma^{-1} (\mathbf{z} - \mathbf{x}')}.$

Let $\mathbf{m} = \Sigma^{-\frac{1}{2}}(\mathbf{z} - \mathbf{x})$ and $\mathbf{u} = \Sigma^{-\frac{1}{2}}(\mathbf{x}' - \mathbf{x})$, then we have $d_{\mathbf{z},\mathbf{x}}^2 = \mathbf{m}^T \mathbf{m}$, $d_{\mathbf{z},\mathbf{x}'}^2 = (\mathbf{m} - \mathbf{u})^T (\mathbf{m} - \mathbf{u})$ and $d_{\mathbf{x}',\mathbf{x}}^2 = \mathbf{u}^T \mathbf{u}$. Moreover, let $\tilde{\mathbf{m}} = -\|\mathbf{m}\|_2 \times \frac{\mathbf{u}}{\|\mathbf{u}\|_2}$, then $\mathbf{m}^T \mathbf{m} = \tilde{\mathbf{m}}^T \tilde{\mathbf{m}}$, and also

$$\begin{aligned} & (\mathbf{m} - \mathbf{u})^T (\mathbf{m} - \mathbf{u}) \\ &= \mathbf{m}^T \mathbf{m} - 2\mathbf{m}^T \mathbf{u} + \mathbf{u}^T \mathbf{u} \\ &\leq \mathbf{m}^T \mathbf{m} + 2\|\mathbf{m}\|_2 \|\mathbf{u}\|_2 + \mathbf{u}^T \mathbf{u} \\ &= (\tilde{\mathbf{m}} - \mathbf{u})^T (\tilde{\mathbf{m}} - \mathbf{u}). \end{aligned} \quad (7)$$

Since $g(\cdot)$ is a monotonically decreasing function, we have

$$\frac{g(\mathbf{m}^T \mathbf{m})}{g[(\mathbf{m} - \mathbf{u})^T (\mathbf{m} - \mathbf{u})]} \leq \frac{g(\tilde{\mathbf{m}}^T \tilde{\mathbf{m}})}{g[(\tilde{\mathbf{m}} - \mathbf{u})^T (\tilde{\mathbf{m}} - \mathbf{u})]}. \quad (8)$$

Considering that $\tilde{\mathbf{m}}^T \tilde{\mathbf{m}} = d_{\mathbf{z},\mathbf{x}}^2$ and $(\tilde{\mathbf{m}} - \mathbf{u})^T (\tilde{\mathbf{m}} - \mathbf{u}) = (d_{\mathbf{z},\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2$, we can conclude that

$$\frac{f(\mathbf{z}|\mathbf{x})}{f(\mathbf{z}|\mathbf{x}')} \leq \frac{g(d_{\mathbf{z},\mathbf{x}}^2)}{g[(d_{\mathbf{z},\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]}, \quad (9)$$

where the equality in (9) holds when $\mathbf{m} = \tilde{\mathbf{m}}$. Since the above derivation relies only on the general properties of elliptical

distributions, rather than any specific form of $g(\cdot)$, the bound in (9) holds for all elliptical distributions considered in this work. Therefore, we obtain the following tight upper bound:

$$\max_z \frac{f(z|\mathbf{x})}{f(z|\mathbf{x}')} = \frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]},$$

Therefore, if $\frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]} \leq e^{\epsilon r}$, then $\frac{f(z|\mathbf{x})}{f(z|\mathbf{x}')} \leq e^{\epsilon r}$. On the other hand, if $\frac{f(z|\mathbf{x})}{f(z|\mathbf{x}')} \leq e^{\epsilon r}$, it implies that $\max_z \frac{f(z|\mathbf{x})}{f(z|\mathbf{x}')} \leq e^{\epsilon r}$ always holds. Then, $\frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]} \leq e^{\epsilon r}$. Thus, we can say, for all $z \in S$, $\frac{f(z|\mathbf{x})}{f(z|\mathbf{x}')} \leq e^{\epsilon r}$ holds if and only if $\frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]} \leq e^{\epsilon r}$ holds. $\square$

Theorem 1 provides the fundamental conditions for determining the feasibility of an elliptical distribution that satisfies ϵ -GI. For the elliptical distributions with PDFs that are continuous at $d_{z,\mathbf{x}} \in [0, +\infty)$, we can further obtain a sufficient condition to directly determine the feasible elliptical distribution for achieving GI.

Corollary 1: For an elliptical distribution with PDF $f(z|\mathbf{x})$, where $f(z|\mathbf{x})$ is continuous for $d_{z,\mathbf{x}} \in [0, +\infty)$, it satisfies ϵ -GI if $\lim_{d_{z,\mathbf{x}} \rightarrow +\infty} \frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]}$ exists, $\forall r > 0$, $\|\mathbf{x} - \mathbf{x}'\|_2 \leq r$, and the upper bound for $\frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]}$ grows with r at an order no greater than $O(e^{\epsilon r})$.

Proof: Since $f(z|\mathbf{x})$ is continuous at $d_{z,\mathbf{x}} \in [0, +\infty)$, both $g(d_{z,\mathbf{x}}^2)$ and $g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]$ are continuous over the same interval. If $\lim_{d_{z,\mathbf{x}} \rightarrow +\infty} \frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]}$ exists, then $g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2] \neq 0$. Consequently, $\frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]}$ is continuous for $d_{z,\mathbf{x}} \in [0, +\infty)$. Since $\lim_{d_{z,\mathbf{x}} \rightarrow +\infty} \frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]}$ exists, there exists $\delta > 0$ such that when $d_{z,\mathbf{x}} > \delta$, $\frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]}$ is upper bounded. Additionally, when $0 \leq d_{z,\mathbf{x}} \leq \delta$, the continuous function $\frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]}$ must be upper bounded. Therefore, we conclude that $\frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]}$ is upper bounded for $d_{z,\mathbf{x}} \in [0, +\infty)$. According to Theorem 1, if the upper bound for $\frac{g(d_{z,\mathbf{x}}^2)}{g[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]}$ grows with r at an order no greater than $O(e^{\epsilon r})$, the elliptical distribution satisfies ϵ -GI. $\square$

B. Feasibility of Elliptical Distributions

As shown in Table I, the PDFs of Laplacian, Gaussian, Student's t , and Logistic distributions are all continuous at $d_{z,\mathbf{x}} \in [0, +\infty)$. Therefore, we can apply Theorem 1 and Corollary 1 to analyze these four typical elliptical distributions.

1) *Planar Laplacian Distribution:* For the planar Laplacian distribution, the function $g_{\mathcal{L}}(d_{z,\mathbf{x}}^2) = e^{-d_{z,\mathbf{x}}}$, then

$$\frac{g_{\mathcal{L}}(d_{z,\mathbf{x}}^2)}{g_{\mathcal{L}}[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]} = \frac{e^{-d_{z,\mathbf{x}}}}{e^{-(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})}} = e^{d_{\mathbf{x}',\mathbf{x}}}. \quad (10)$$

Note that

$$d_{\mathbf{x}',\mathbf{x}} = \sqrt{(\mathbf{x}' - \mathbf{x})^T \Sigma^{-1} (\mathbf{x}' - \mathbf{x})} \leq \frac{r}{\sqrt{\lambda_{\min}}}, \quad (11)$$

where $\lambda_{\min}$ represents the smallest eigenvalue of Σ , and the equality holds when Σ has two identical eigenvalues. According to Theorem 1, the planar Laplacian distribution is feasible for achieving ϵ -GI.

2) *Gaussian Distribution:* For the Gaussian distribution, the function $g_{\mathcal{N}}(d_{z,\mathbf{x}}^2) = e^{-\frac{1}{2}d_{z,\mathbf{x}}^2}$. Then we have

$$\frac{g_{\mathcal{N}}(d_{z,\mathbf{x}}^2)}{g_{\mathcal{N}}[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]} = \frac{e^{-\frac{1}{2}d_{z,\mathbf{x}}^2}}{e^{-\frac{1}{2}(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2}} = e^{d_{z,\mathbf{x}}d_{\mathbf{x}',\mathbf{x}} + \frac{1}{2}d_{\mathbf{x}',\mathbf{x}}^2}. \quad (12)$$

The ratio in (12) increases with $d_{z,\mathbf{x}}$ and $d_{\mathbf{x}',\mathbf{x}}$, and for any given r , it tends to infinity as $d_{z,\mathbf{x}} \rightarrow \infty$. Consequently, the Gaussian distribution does not satisfy the condition in Theorem 1, confirming that it is infeasible for achieving ϵ -GI.

3) *Student's t Distribution:* For the Student's t distribution, the function $g_{\mathcal{T}}(d_{z,\mathbf{x}}^2) = (1 + \frac{1}{v}d_{z,\mathbf{x}}^2)^{-\frac{v+2}{2}}$. Then we have

$$\frac{g_{\mathcal{T}}(d_{z,\mathbf{x}}^2)}{g_{\mathcal{T}}[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]} = \left[\frac{1 + \frac{1}{v}(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2}{1 + \frac{1}{v}d_{z,\mathbf{x}}^2} \right]^{\frac{v+2}{2}}. \quad (13)$$

When $d_{z,\mathbf{x}} \rightarrow +\infty$, we have

$$\lim_{d_{z,\mathbf{x}} \rightarrow +\infty} \frac{g_{\mathcal{T}}(d_{z,\mathbf{x}}^2)}{g_{\mathcal{T}}[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]} = 1. \quad (14)$$

Moreover, according to (11), the upper bound of ratio in (13) is a polynomial function of r . Therefore, the Student's t distribution satisfies the condition in Corollary 1, confirming that it is feasible for achieving ϵ -GI.

4) *Logistic Distribution:* For the Logistic distribution, the function $g_{\mathcal{C}}(d_{z,\mathbf{x}}^2) = \frac{e^{-\frac{1}{2}d_{z,\mathbf{x}}^2}}{(1 + e^{-\frac{1}{2}d_{z,\mathbf{x}}^2})^2}$. Then we have

$$\frac{g_{\mathcal{C}}(d_{z,\mathbf{x}}^2)}{g_{\mathcal{C}}[(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]} = e^{d_{z,\mathbf{x}}d_{\mathbf{x}',\mathbf{x}} + \frac{1}{2}d_{\mathbf{x}',\mathbf{x}}^2} \times \left[\frac{e^{-\frac{1}{2}(d_{z,\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2} + 1}{e^{-\frac{1}{2}d_{z,\mathbf{x}}^2} + 1} \right]^2. \quad (15)$$

In the ratio given by (15), one term increases with $d_{z,\mathbf{x}}$ and $d_{\mathbf{x}',\mathbf{x}}$, while the other remains upper bounded by 1. For any given r , this ratio tends to infinity as $d_{z,\mathbf{x}} \rightarrow \infty$. Consequently, the Logistic distribution fails to satisfy the condition in Theorem 1, which implies that it is infeasible to achieve ϵ -GI.

V. PRIVACY LEVEL OF STUDENT'S t DISTRIBUTION

For Student's t distribution, the quantitative relationship between the parameters of its PDF and the privacy level remains unknown. In the following analysis, we will establish the quantitative relationship between the privacy level and the parameters of the Student's t distribution.

To compute the privacy level, we need to derive the upper bound of $\frac{P_T(S|\mathbf{x})}{P_T(S|\mathbf{x}')}$. We first derive the upper bound of $\frac{f_T(\mathbf{z}|\mathbf{x})}{f_T(\mathbf{z}|\mathbf{x}')}$. It follows from (9) that

$$\frac{f_T(\mathbf{z}|\mathbf{x})}{f_T(\mathbf{z}|\mathbf{x}')} \leq \frac{g_T(d_{\mathbf{z},\mathbf{x}}^2)}{g_T[(d_{\mathbf{z},\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]} = \left[\frac{v + (d_{\mathbf{z},\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2}{v + d_{\mathbf{z},\mathbf{x}}^2} \right]^{\frac{v+2}{2}}. \quad (16)$$

According to (7), (8) and (9), we conclude that the first equality in (16) holds when

$$\Sigma^{-\frac{1}{2}}(\mathbf{z} - \mathbf{x}) = - \left\| \Sigma^{-\frac{1}{2}}(\mathbf{z} - \mathbf{x}) \right\|_2 \times \frac{\Sigma^{-\frac{1}{2}}(\mathbf{x}' - \mathbf{x})}{\left\| \Sigma^{-\frac{1}{2}}(\mathbf{x}' - \mathbf{x}) \right\|_2}.$$

The ratio $\frac{g_T(d_{\mathbf{z},\mathbf{x}}^2)}{g_T[(d_{\mathbf{z},\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]}$ is first increasing and then decreasing with respect to $d_{\mathbf{z},\mathbf{x}}$, and its maximum value is achieved at the stationary point

$$d_{\mathbf{z},\mathbf{x}}^* = \sqrt{v + \frac{d_{\mathbf{x}',\mathbf{x}}^2}{4}} - \frac{d_{\mathbf{x}',\mathbf{x}}}{2}, \quad (17)$$

where the first-order derivative equals zero. Thus, the maximum value is given by:

$$\frac{g_T(d_{\mathbf{z},\mathbf{x}}^{*2})}{g_T[(d_{\mathbf{z},\mathbf{x}}^* + d_{\mathbf{x}',\mathbf{x}})^2]} = \left(1 + \frac{2}{\sqrt{\frac{4v}{d_{\mathbf{x}',\mathbf{x}}^2} + 1} - 1} \right)^{\frac{v+2}{2}}. \quad (18)$$

It is obvious that the above function increases monotonically with respect to $d_{\mathbf{x}',\mathbf{x}}^2$. Substituting (11) into (18), we obtain:

$$\frac{g_T(d_{\mathbf{z},\mathbf{x}}^{*2})}{g_T[(d_{\mathbf{z},\mathbf{x}}^* + d_{\mathbf{x}',\mathbf{x}})^2]} \leq \left(1 + \frac{2}{\sqrt{\frac{4v\lambda_{\min}}{r^2} + 1} - 1} \right)^{\frac{v+2}{2}}. \quad (19)$$

Therefore, we have

$$\frac{f_T(\mathbf{z}|\mathbf{x})}{f_T(\mathbf{z}|\mathbf{x}')} \leq \left(1 + \frac{2}{\sqrt{\frac{4v\lambda_{\min}}{r^2} + 1} - 1} \right)^{\frac{v+2}{2}}. \quad (20)$$

Consequently, we have

$$\frac{P_T(S|\mathbf{x})}{P_T(S|\mathbf{x}')} = \frac{\int_S f_T(\mathbf{z}|\mathbf{x}) d\mathbf{z}}{\int_S f_T(\mathbf{z}|\mathbf{x}') d\mathbf{z}} \leq \left(1 + \frac{2}{\sqrt{\frac{4v\lambda_{\min}}{r^2} + 1} - 1} \right)^{\frac{v+2}{2}}, \quad (21)$$

where the equality in (21) holds when $d_{\mathbf{x}',\mathbf{x}}^2 = \frac{r^2}{\lambda_{\min}}$, $d_{\mathbf{z},\mathbf{x}} = d_{\mathbf{z},\mathbf{x}}^*$ and $\frac{f_T(\mathbf{z}|\mathbf{x})}{f_T(\mathbf{z}|\mathbf{x}')} = \frac{g_T(d_{\mathbf{z},\mathbf{x}}^2)}{g_T[(d_{\mathbf{z},\mathbf{x}} + d_{\mathbf{x}',\mathbf{x}})^2]}$.

According to Definition 2, we ensure that the Student's t distribution satisfies ϵ -GI by setting

$$e^{\epsilon r} = \left(1 + \frac{2}{\sqrt{\frac{4v\lambda_{\min}}{r^2} + 1} - 1} \right)^{\frac{v+2}{2}}. \quad (22)$$

Therefore, the privacy level guaranteed by the Student's t distribution is given by:

$$\ell_T = \epsilon r = \frac{v+2}{2} \ln \left(1 + \frac{2}{\sqrt{\frac{4v\lambda_{\min}}{r^2} + 1} - 1} \right). \quad (23)$$

The above equation illustrates the quantitative relationship between the privacy level and Student's t distribution's parameters: the DoF v , the scale matrix Σ and the maximum distance between $\mathbf{x}$ and $\mathbf{x}'$, i.e., r . For given v , Σ , and r , the value of ϵ remains constant. The privacy level derived in (23) increases with r , which is consistent with the definition of GI. This behavior ensures a stronger privacy guarantee near the user's location while allowing for higher distinguishability at more distant points, with given v , Σ , and ϵ .

Remark 2: The DoF v and the scale matrix Σ affect the mechanism in different ways. The DoF v determines the shape of the distribution, with smaller values of v produce heavier tails and make large perturbations more likely. The scale matrix Σ determines the perturbation magnitude via its eigenvalues, where larger eigenvalues result in larger perturbations. These effects are consistent with the privacy level in (23) and the data utility in (24). Specifically, with $\lambda_{\min}$ fixed, a larger v weakens the privacy guarantee while improving the data utility. With v fixed, a larger $\lambda_{\min}$ strengthens the privacy guarantee while reducing the data utility. Therefore, when using the Student's t distribution for GI, it is necessary to select (v, Σ) appropriately to balance the privacy level and data utility in practical LBS applications.

VI. OPTIMAL MECHANISM DESIGN

Similar to the planar Laplacian mechanism, the randomization mechanism based on Student's t distribution inevitably degrades the data utility. The tradeoff between privacy preservation and data utility is necessary. In this section, we aim to determine the values of DoF and the scale matrix of Student's t distribution to balance the data utility and privacy level. We formulate two optimization problems in the randomization mechanism for two different scenarios.

A. Scenario 1: Maximal Data Utility

According to (4), the data utility under Student's t distribution is the inverse of the trace of the covariance matrix:

$$u_T = \frac{v-2}{v} \cdot \frac{1}{\text{tr}(\Sigma)} = \frac{v-2}{v} \cdot \frac{1}{\lambda_{\min} + \lambda_{\max}}. \quad (24)$$

A higher value of u_T indicates better data utility.

In the first scenario, the user has a minimum requirement for location privacy and wants to maximize the data utility. For example, users may contribute their private locations to mobile crowdsensing or third-party analytics tasks, including population distribution estimation [25], traffic monitoring [26], and environmental monitoring [55]. Therefore, we construct the optimization problem Q1 that maximizes the data utility u_T under the constraint of privacy level. The formulation of Q1 is

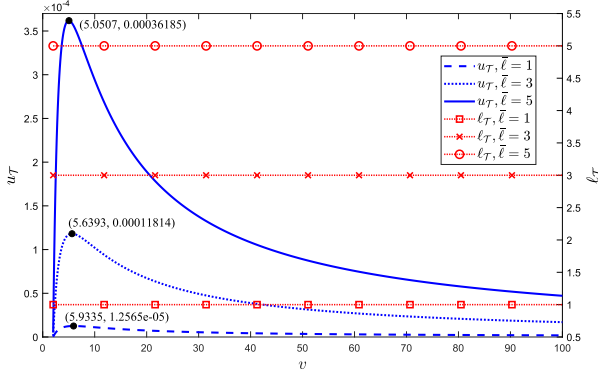


Fig. 1. Scenario 1: The data utility $u_{\mathcal{T}}$ and privacy level $\ell_{\mathcal{T}}$ under different values of v .

as follows:

$$\begin{aligned} Q1 : \max_{v, \Sigma} u_{\mathcal{T}}, \\ \text{s.t. } \ell_{\mathcal{T}} \leq \bar{\ell}, \\ v > 2, \end{aligned} \quad (25)$$

where $\bar{\ell}$ represents the upper bound for privacy level. Recall that the lower the privacy level is, the stronger the privacy guarantee the user obtains. The optimal values for v and Σ are given by the following theorem.

Theorem 2: Given a privacy level threshold $\bar{\ell}$, the data utility of Student's t distribution is maximized when:

- 1) The scale matrix Σ has two identical eigenvalues, which is given by

$$\lambda_1(v_1^*) = \frac{r^2}{v_1^* \left(e^{\frac{2\bar{\ell}}{v_1^{*+2}}} + e^{-\frac{2\bar{\ell}}{v_1^{*+2}}} - 2 \right)}, \quad (26)$$

where v_1^* is the optimal DoF.

- 2) The optimal DoF v_1^* is at the stationary point of $u'_{\mathcal{T}}(v)$, and

$$u'_{\mathcal{T}}(v) = \frac{v-2}{2r^2} \left(e^{\frac{2\bar{\ell}}{v+2}} + e^{-\frac{2\bar{\ell}}{v+2}} - 2 \right), \quad (27)$$

where $2 < v_1^* < 6$.

Theorem 2 provides sufficient conditions for the optimal solution of Q_1 . As long as the scale matrix and DoF of Student's t distribution meet the conditions in Theorem 2, the randomization mechanism based on Student's t distribution can always provide the highest data utility under the constraint of privacy level. To be more intuitive, we show the data utility $u_{\mathcal{T}}$ and privacy level $\ell_{\mathcal{T}}$ under different values of v in Fig. 1. In this figure, we set $r = 100$ m, the scale matrix is set as a diagonal matrix with diagonal elements being $\lambda_1(v)$, the resulting data utility $u_{\mathcal{T}} = u'_{\mathcal{T}}(v)$, and the privacy level $\ell_{\mathcal{T}}$ is obtained by (23). As shown in Fig. 1, the data utility $u_{\mathcal{T}}$ has a unique stationary point when v is in the interval (2,6). When $v \geq 6$, $u_{\mathcal{T}}$ is decreasing with v . The data utility $u_{\mathcal{T}}$ always reaches its maximum value at the stationary point within the interval (2,6). Regarding $\ell_{\mathcal{T}}$, it is always equivalent to $\bar{\ell}$ regardless of the value of v . The

reason is that $\ell_{\mathcal{T}} = \bar{\ell}$ always holds when the scale matrix has two identical eigenvalues. Therefore, the privacy level constraint in $Q1$ always holds. These results validate the discussion in the proof of Theorem 2.

B. Scenario 2: Optimal Privacy Preservation

In the second scenario, the user needs to guarantee data utility above a certain threshold to maintain service quality. For example, the user relies on perturbed locations to obtain services, including indoor localization [56], [57], points-of-interest queries [58], and ride-hailing service [59]. We formulate the optimization problem $Q2$ to minimize the privacy level, i.e., maximize the privacy guarantee, under the constraint of data utility. The optimization problem is as follows:

$$\begin{aligned} Q2 : \min_{v, \Sigma} \ell_{\mathcal{T}}, \\ \text{s.t. } u_{\mathcal{T}} \geq \underline{u}, \\ v > 2, \end{aligned} \quad (28)$$

where $\underline{u}$ represents the minimum expected data utility. The solutions of $Q2$ are summarized in the following theorem.

Theorem 3: Under the constraint of data utility being no less than $\underline{u}$, the privacy guarantee provided by Student's t distribution is maximized when

- 1) The scale matrix Σ has two identical eigenvalues, i.e.,

$$\lambda_2(v_2^*) = \frac{v_2^* - 2}{2v_2^*} \cdot \frac{1}{\underline{u}}, \quad (29)$$

where v_2^* is the optimal DoF.

- 2) The optimal DoF v_2^* is the stationary point of $\ell'_{\mathcal{T}}(v)$, where $2 < v_2^* < 14$ and $\ell'_{\mathcal{T}}(v)$ is given by

$$\ell'_{\mathcal{T}}(v) = \frac{v+2}{2} \ln \left(1 + \frac{2}{\sqrt{\frac{2(v-2)}{ur^2} + 1} - 1} \right). \quad (30)$$

Theorem 3 provides sufficient conditions for the optimal solution of Q_2 . As long as the scale matrix and DoF of Student's t distribution meet the conditions in Theorem 3, the randomization mechanism based on Student's t distribution can always provide the strongest privacy guarantee under the constraint of data utility. To be more intuitive, we show the privacy level $\ell_{\mathcal{T}}$ and data utility $u_{\mathcal{T}}$ under different values of v in Fig. 2. In this figure, we set $r = 100$ m, the scale matrix is set as a diagonal matrix with diagonal elements being $\lambda_2(v)$ in (29), the resulting privacy level $\ell_{\mathcal{T}} = \ell'_{\mathcal{T}}(v)$, and the data utility $u_{\mathcal{T}}$ is obtained by (24). As shown in Fig. 2, $\ell_{\mathcal{T}}$ exhibits convex behavior when $2 < v < \bar{v}$ and becomes concave for $v \geq \bar{v}$. Moreover, for $v \geq \bar{v}$, $\ell_{\mathcal{T}}$ increases with v . Under the constraint of data utility $\underline{u}$, the minimal value of $\ell_{\mathcal{T}}$ is achieved at its stationary point. Regarding $u_{\mathcal{T}}$, it is always equivalent to $\underline{u}$ regardless of the value of v . The reason is that $u_{\mathcal{T}} = \underline{u}$ always holds when the scale matrix has two identical eigenvalues. Therefore, the data utility constraint in $Q2$ always holds. These findings again validate the analysis in the proof of Theorem 3.

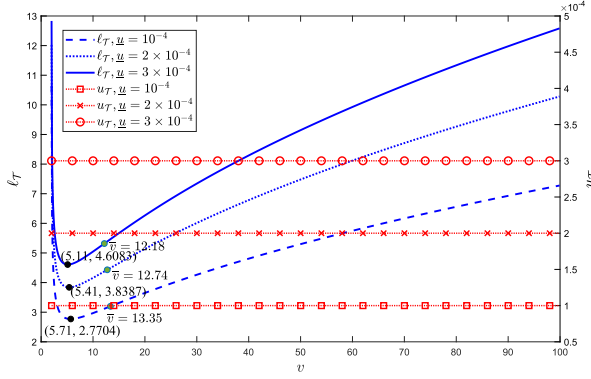


Fig. 2. Scenario 2: The data utility u_T and privacy level ℓ_T under different values of v .

C. Implementation and Complexity Analysis

In summary, implementing GI with the Student's t distribution involves two steps: optimal parameter selection and perturbation generation. Given a privacy constraint or a utility constraint, the optimal parameters (v^*, Σ^*) can be determined according to Theorem 2 or Theorem 3, respectively. Under the optimal solution, Σ^* has two identical eigenvalues, denoted by λ^* . Then, an angle $\theta \sim \mathcal{U}(0, 2\pi)$ is sampled, and the perturbed location is generated as

$$\tilde{\mathbf{x}} = \mathbf{x} + \tilde{r} [\cos(\theta), \sin(\theta)]^\top, \quad (31)$$

where $\tilde{r}$ denotes the perturbation radius. Let $\alpha \sim \mathcal{U}(0, 1)$. By applying inverse transform sampling to the radial CDF of the Student's t -based mechanism, $\tilde{r}$ is given by

$$\tilde{r} = \sqrt{\frac{v^*}{\lambda^*} ((1 - \alpha)^{-2/v^*} - 1)}. \quad (32)$$

The computational complexity of the proposed mechanism consists of two parts: the computation of (v^*, Σ^*) and the generation of $\tilde{\mathbf{x}}$. In both scenarios, computing v^* reduces to solving a one-dimensional stationary equation, using (27) in Scenario 1 and (30) in Scenario 2, which requires $\mathcal{O}(T)$ complexity when Newton's method is applied with T iterations. The closed-form computation of λ^* requires $\mathcal{O}(1)$. For perturbation generation, sampling α and θ , computing $\tilde{r}$, and performing the coordinate transformation all require only constant-time operations, resulting in $\mathcal{O}(1)$ complexity. Therefore, if the privacy or utility threshold remains the same for the whole dataset, the optimal parameters need to be computed only once and can then be reused for all N locations, leading to a total complexity of $\mathcal{O}(T + N)$. If the threshold depends on the location, the parameter selection step must be performed separately for each location, resulting in a total complexity of $\mathcal{O}(TN)$. This demonstrates that the proposed mechanism is computationally efficient and suitable for large-scale location datasets.

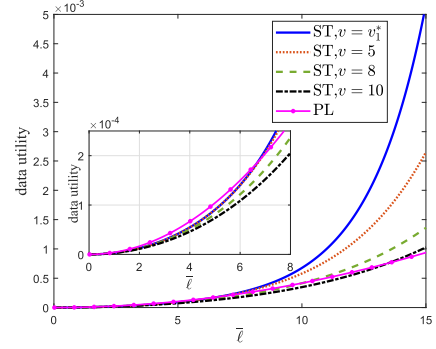


Fig. 3. The comparison of data utility between ST and PL under the constraint of privacy level.

VII. PERFORMANCE EVALUATION

In this section, we compare the performance of Student's t distribution with the widely applied planar Laplacian mechanism [10] in terms of data utility and location privacy. Then, we apply the randomization mechanisms to two typical location-based applications, including fingerprint-based indoor localization and location-based advertising, and evaluate the corresponding system performance. In the experiments, the Student t mechanism and planar Laplacian mechanism are respectively denoted by ST and PL for short.

A. Comparison With Planar Laplacian Mechanism

In this part, we let the actual location $\mathbf{x} = [0, 0]^T$ and $r = 200$ m. The randomization mechanism ST or PL generates the perturbed locations $\mathbf{z}$. We compare the performance of ST and PL in different scenarios.

1) *Data Utility Comparison in Scenario 1:* In this scenario, we compare the data utility of ST and PL under the constraint of privacy level. The optimal parameters of ST can be determined by Theorem 2. The privacy parameter ϵ of Student's t and planar Laplacian mechanisms is the same. For a given r , the optimal value of ϵ to maximize the data utility is $\epsilon_S^* = \epsilon_L^* = \frac{\bar{\ell}}{r}$. For PL, according to (4), the data utility function can be expressed as $u_L = \frac{\epsilon^2}{6}$, and the maximum value of data utility is

$$u_L^* = \frac{\bar{\ell}^2}{6r^2} \quad (33)$$

Data utilities of ST and PL under different privacy level constraints are shown in Fig. 3. Regarding ST, we can see that the optimal DoF v_1^* brings the highest data utility compared to other values of v . Compared with PL, we can see that ST does not always outperform PL. When $\bar{\ell}$ is small, PL slightly outperforms ST. When $\bar{\ell}$ exceeds some value, ST provides higher data utility, and the gap between ST and PL becomes larger as $\bar{\ell}$ increases. Considering that under a given value of $\bar{\ell}$, according to Theorem 2 and (33), we can easily obtain the maximum values of data utility provided by ST and PL. In practical application, we can select the best mechanism that provides higher data utility according to the value of $\bar{\ell}$.

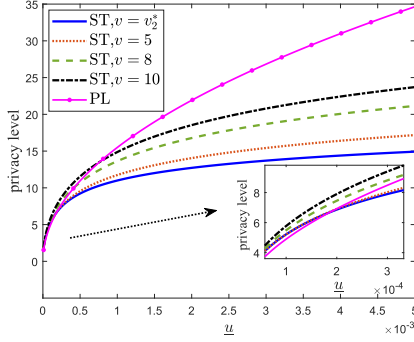


Fig. 4. The comparison of privacy level between ST and PL under the constraint of data utility.

2) *Privacy Level Comparison in Scenario 2:* In this scenario, we compare the privacy level of ST and PL under the constraint of data utility. The optimal parameters of ST can be determined by Theorem 3. The privacy parameter ϵ of the Student's t mechanism is determined by submitting the optimal DoF v_2^* into (23). Since the data utility function of PL is $u_{\mathcal{L}} = \frac{\epsilon^2}{6}$, for a given r , the optimal value of ϵ to minimize the privacy level is $\epsilon_{\mathcal{L}}^* = \sqrt{6u}$, and the minimum value of privacy level is given by

$$\ell_{\mathcal{L}}^* = \sqrt{6ur} \quad (34)$$

The privacy levels of ST and PL under different data utility constraints are illustrated in Fig. 4. Regarding ST, we observe that the optimal DoF v_2^* guarantees the strongest privacy preservation compared to other values of v . In comparison with PL, similar to Scenario 1, ST does not always outperform PL. In the case of low data utility, PL slightly provides a stronger privacy guarantee than ST. In the case of high data utility, ST provides a stronger privacy guarantee, and the difference between ST and PL becomes more pronounced as u increases. Under a given value of u , according to Theorem 3 and (34), we can easily determine the minimum values of privacy level provided by ST and PL. In practical application, we can select the best mechanism that provides a stronger privacy guarantee according to the value of u .

3) *Privacy Loss Comparison:* GI provides a statistical evaluation of the level of location privacy preservation. Apart from that, we also evaluate the actual privacy loss for a certain perturbed location. Under the same privacy level, the actual privacy loss of different perturbed locations may differ. According to [31], the privacy loss is defined as

$$\ell_{\mathcal{M}, \mathbf{x}, \mathbf{x}'}(\mathbf{z}) = \left| \ln \left(\frac{f(\mathbf{z}|\mathbf{x})}{f(\mathbf{z}|\mathbf{x}')} \right) \right| \quad (35)$$

We can see that the privacy loss caused by reporting $\mathbf{z}$ is associated with the locations $\mathbf{x}$, $\mathbf{x}'$ and the privacy mechanism $\mathcal{M}$. Different from the privacy level defined in GI, privacy loss evaluates the actual geographic indistinguishability for a certain perturbed location generated by the randomization mechanism. The upper bound of privacy loss is the defined privacy level.

The privacy loss difference between ST and PL is defined as $\Delta \ell_{\mathcal{M}, \mathbf{x}, \mathbf{x}'}(\mathbf{z}) = \ell_{\mathcal{T}, \mathbf{x}, \mathbf{x}'}(\mathbf{z}) - \ell_{\mathcal{L}, \mathbf{x}, \mathbf{x}'}(\mathbf{z})$, where $\ell_{\mathcal{T}, \mathbf{x}, \mathbf{x}'}(\mathbf{z})$ and

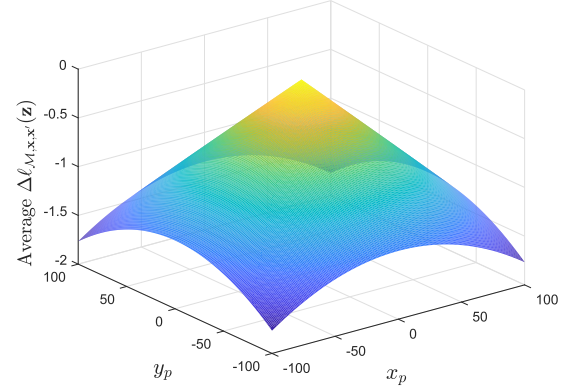


Fig. 5. The average difference between the privacy loss of ST and PL.

$\ell_{\mathcal{L}, \mathbf{x}, \mathbf{x}'}(\mathbf{z})$ represent the privacy loss of ST and PL, respectively. Let $\ell = 8$, $\epsilon = 0.04$, $v = 5$, $\mathbf{x}' = [x_p, y_p]^T$, and $\|\mathbf{x} - \mathbf{x}'\|_2 \leq r$. Fig. 5 illustrates the average difference between the privacy loss of ST and PL across 1000^2 perturbed locations in a $1000 \text{ m} \times 1000 \text{ m}$ square area. As shown in Fig. 5, the averaged privacy loss differences are all negative, which verifies that ST leads to lower privacy loss than PL, even when both mechanisms guarantee the same privacy level.

B. Fingerprint-Based Indoor Localization

Fingerprint-based localization has become one of the most widely used methods in indoor localization due to its simplicity and efficiency [60]. Typically, the fingerprint-based localization consists of an offline phase and an online phase. In the offline phase, location coordinates and the corresponding location-dependent fingerprints, e.g., received signal strength (RSS), are collected at reference points, which construct a fingerprint database. During the online phase, users collect real-time measurements and match them with the fingerprints stored in the offline database via specific algorithms, e.g., KNN or machine learning methods, for location estimation. However, collecting location information in the offline phase raises the risk of users' privacy leakage. Location privacy preservation in the offline phase is essential. In this part, we evaluate the localization accuracy when we apply a randomization mechanism to protect the location data in the fingerprint database.

We conduct experiments on a popular fingerprint database called UJIIndoorLoc [61]. It contains 21049 WiFi RSS fingerprint samples at 933 reference points in 3 buildings of the Jaume I University. In our experiment, we evaluate the localization performance in Floor 1 of Building 1, where the offline fingerprint database contains 1484 samples and the validation set contains 143 samples. To protect the location privacy, we perturb the location coordinates within the fingerprint database by the randomization mechanism (ST and PL). The Weighted k -Nearest Neighbor algorithm with $k = 3$ is employed to estimate the location. The Euclidean distance between estimated and exact locations is used to measure the localization error.

We first compare both the localization performance and the data utility under different randomization mechanisms with

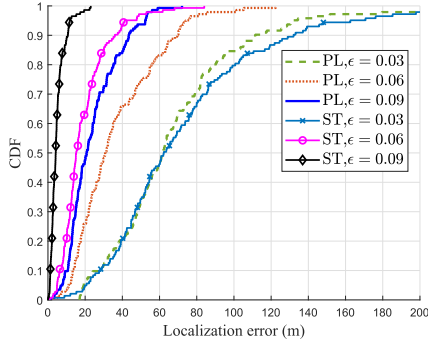


Fig. 6. CDF of localization error under different randomization mechanisms with different privacy levels.

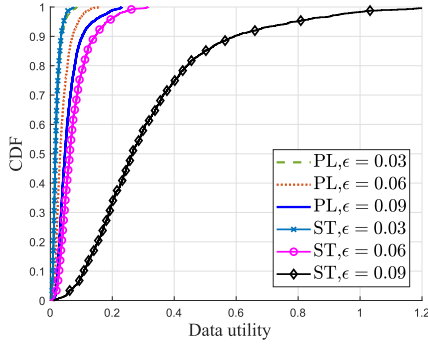
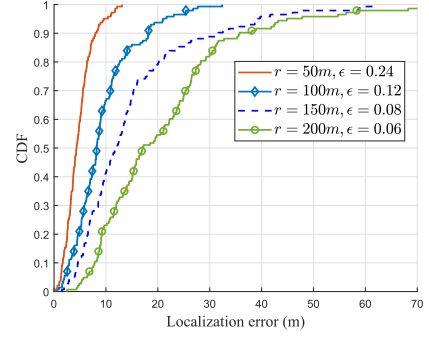


Fig. 7. CDF of data utility in indoor localization under different randomization mechanisms and privacy levels.

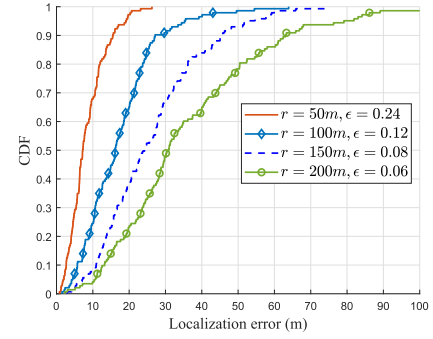
different privacy levels. Let $r = 200$ m and vary the value of ϵ , the corresponding privacy level changes. Figs. 6 illustrates the cumulative distribution function (CDF) of the localization errors under ST and PL. It can be observed that the localization error decreases as the privacy level increases. This observation is reasonable since a higher privacy level corresponds to a weaker privacy guarantee, consequently implying a higher data utility. When $\epsilon = 0.03$, the localization error under ST is slightly higher than that under PL, although both privacy mechanisms result in higher localization errors. When ϵ becomes larger, e.g., $\epsilon = 0.06, 0.09$, the localization performance under ST is much better than that under PL.

Fig. 7 illustrates the CDF of the data utility under ST and PL. The Student's t mechanism (ST) achieves higher data utility than the planar Laplacian mechanism (PL) at larger values of ϵ (e.g., $\epsilon = 0.06, 0.09$) while maintaining comparable utility when $\epsilon = 0.03$. The findings in Figs. 6 and 7 are consistent with the results in Section VII-A: the data utility of PL is slightly higher than that of ST at lower privacy levels but much lower at higher privacy levels.

We then evaluate the influence of r on localization performance while maintaining a fixed privacy level $\bar{\ell} = 12$. Fig. 8 presents the CDF of the localization errors under both ST and PL for different values of r , keeping $\bar{\ell}$ unchanged. We observe that the localization error degrades as r decreases, which is consistent with the theoretical result in (27). According to (27), data utility is inversely proportional to r^2 when $\bar{\ell}$ is fixed. The same results can be found under PL, as shown in Fig. 8(b).



(a) ST



(b) PL

Fig. 8. CDF of localization error under ST and PL with different r but the same privacy level $\bar{\ell}$.

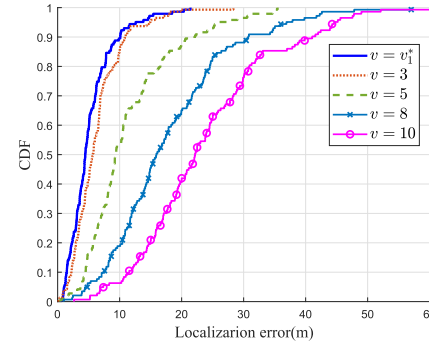


Fig. 9. CDF of localization error under ST with different v .

Next, we validate that ST can achieve the best localization performance with the optimal DoF v_1^* . We set $r = 200$ m and $\epsilon = 0.09$. Fig. 9 illustrates the CDF of localization error under ST with different values of v . The result is consistent with that in Theorem 2. The optimal value of v maximizes the data utility, and the corresponding localization performance would be the best under ST.

C. Location-Based Advertising

Location-based advertising (LBA) is a typical location-based service; it provides location-specific advertisements to users. The service quality of LBA relies on accurate locations, whereas

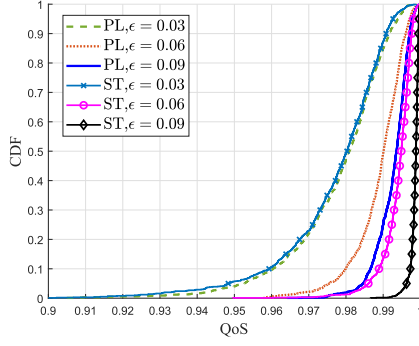


Fig. 10. CDF of QoS under different randomization mechanisms with different privacy levels.

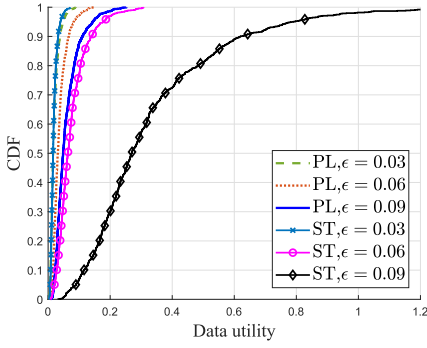


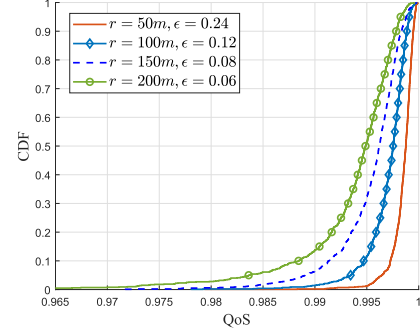
Fig. 11. CDF of data utility in LBA under different randomization mechanisms with different privacy levels.

user location collection raises privacy leakage concerns. In this part, we evaluate the quality of service (QoS) when we apply the randomization mechanism to protect users' location data. Suppose a user is actually located at $\mathbf{x}$, he requests points of interest recommendation from the server within a radius of r_{LBA} by reporting a perturbed location $\mathbf{z}$. As defined in [62], the QoS is evaluated by the overlapping region area between two circles centered at $\mathbf{x}$ and $\mathbf{z}$ respectively, each with a radius of r_{LBA} . It is computed as

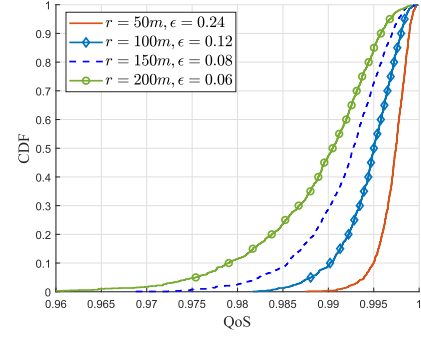
$$\text{QoS} = \frac{2 \arccos\left(\frac{r_g}{2r_{LBA}}\right)}{\pi} - \frac{r_g \sqrt{1 - \left(\frac{r_g}{2r_{LBA}}\right)^2}}{\pi r_{LBA}} \quad (36)$$

where $r_g = \|\mathbf{x} - \mathbf{z}\|_2$ and r_{LBA} is the service radius defined by the user or the service provider. Let $r_{LBA} = 2000$ m and $\mathbf{x} = [0, 0]^T$. In each experiment, we generate 1000 perturbed locations by each randomization mechanism.

We first compare the QoS and data utility under different randomization mechanisms with different privacy levels. Let $r = 200$ m and vary the value of ϵ , the corresponding privacy level changes. Fig. 10 and 11 respectively show the CDF of the QoS and data utility under ST and PL with different privacy levels. We can see that, for each randomization mechanism, a larger ϵ leads to a higher QoS and better data utility. Compared with PL, ST produces higher QoS and data utility at larger values of ϵ , e.g., $\epsilon = 0.06, 0.09$, but slightly lower QoS when $\epsilon = 0.03$. These findings are consistent with the results in Figs. 3, 6, and 7.



(a) ST



(b) PL

Fig. 12. CDF of QoS under ST and PL with different r but the same privacy level $\bar{\ell}$.

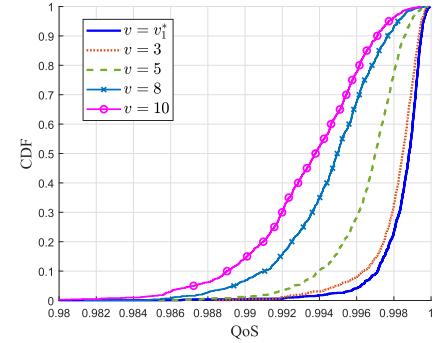


Fig. 13. CDF of QoS under ST with different v .

We then evaluate the influence of r on QoS while maintaining a fixed privacy level $\bar{\ell} = 12$. Fig. 12 shows the CDF of the QoS under ST and PL for different values of r while keeping $\bar{\ell}$ unchanged. We can see that a larger r leads to a higher degradation in QoS, which again verifies that the data utility of randomization mechanisms increases as r decreases when $\bar{\ell}$ is fixed.

Next, we validate that ST can achieve the best QoS with the optimal DoF v_1^* under the privacy level constraint. We set $r = 200$ m and $\epsilon = 0.09$. Fig. 13 presents the CDF of QoS under ST with different values of v . It is observed that the QoS is

maximized when v equals v_1^* . These findings are also consistent with Theorem 2.

VIII. DISCUSSION AND FUTURE WORKS

A. Data Utility of GI Randomization Mechanisms

The Student's t mechanism exhibits heavier tails compared to the planar Laplacian mechanism. Although the experimental results demonstrate that it improves data utility compared to the planar Laplacian mechanism at higher privacy levels, we cannot directly conclude that GI randomization mechanisms with heavier tails yield better data utility, as the investigation into GI randomization mechanisms remains underdeveloped.

To further explore this issue, we analyze the tail properties of state-of-the-art DP mechanisms and their impact on data utility. The offset-symmetric Gaussian tail distribution proposed in [39] for (ϵ, δ) -DP exhibits heavier tails than the Gaussian distribution and achieves higher data utility at the same privacy level. However, another study highlights that tail properties alone are not enough to determine data utility. As discussed in [40], both tail properties and normalized Fisher information (NFI) are critical. Despite having lighter tails than the Gaussian distribution, the flipped Huber mechanism proposed in [40] achieves better data utility at the same privacy level due to its higher NFI. These findings indicate that while tail properties are important for the data utility of GI mechanisms, other factors, such as NFI, can also play a critical role. This suggests the need for further investigation into the relationship between the distribution characteristics of GI randomization mechanisms and their impact on data utility.

B. Extensions to Broader LBS Scenarios

1) *Continuous LBS Queries*: In continuous LBS queries, independently generating perturbed locations for multiple queries ensures $n\epsilon$ -GI for n queries if each query uses a randomization mechanism satisfying ϵ -GI. Intuitively, this approach leads to privacy degradation as the number of queries increases [49]. To improve the privacy guarantee, previous methods have proposed reconstructing reported locations from perturbed ones [63], [64], or dynamically adjusting the privacy parameter ϵ [51], [65]. These approaches rely on the planar Laplacian mechanism to generate perturbed locations, and exploring alternative GI randomization mechanisms could enhance these methods. In this work, we propose the Student's t mechanism for achieving GI. It performs much better than the planar Laplacian mechanism at higher privacy levels, offering promising improvements in privacy guarantees for continuous LBS query scenarios. By expanding GI randomization mechanisms, our work provides guidance for future research on continuous LBS queries within the GI framework.

2) *3D LBS Scenarios*: Since indoor locations are naturally represented in three dimensions, privacy preservation in three-dimensional (3D) space is also of practical interest. Min et al. introduced the privacy notion of 3D-GI for indoor LBS, where the privacy guarantee is defined over 3D coordinates and a three-variate Laplacian perturbation mechanism is designed accordingly [7], [66]. Formally, 3D-GI extends the GI notion

from planar locations to 3D coordinates by defining the privacy guarantee with the 3D Euclidean distance. From this perspective, it may be possible to extend the Student's t mechanism to the 3D-GI setting, since the multivariate Student's t distribution is well defined in higher-dimensional spaces. However, additional analysis is still needed to verify whether the resulting mechanism satisfies 3D-GI. In future work, we will further investigate the feasibility of using the multivariate Student's t distribution to achieve 3D-GI and evaluate its privacy and utility performance.

IX. CONCLUSION

In this paper, we investigate elliptical distributions for the randomization mechanisms to achieve GI. To evaluate the feasibility of elliptical distributions, we provide two necessary and sufficient conditions for achieving GI, through which we show that, except for the planar Laplacian distribution, the Student's t distribution is also feasible for GI. Then, we establish the quantitative relationship between the privacy level in GI and the key parameters of Student's t distribution. Furthermore, to balance the data utility and privacy level, we design the optimal randomization mechanisms under the Student's t distribution in two scenarios. The first scenario aims to maximize the data utility under the constraint of privacy level, while the second scenario aims to minimize the privacy level under the constraint of data utility. Closed-form solutions for these two scenarios are both derived. Finally, extensive numerical results validate that the Student's t mechanism performs much better than the commonly used planar Laplacian mechanism at higher privacy levels.

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